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THESE P AND I DIAGRAM IS DIAGRAMATIC ONLY. DESIGN OF PIPE LINES MUST BE INVESTIGATED FOR VENTING OF GAS AND VAPOR POCKETS IN PIPING AND EQUIPMENT. LOW POINTS IN PIPING, PUMPS AND EQUIPMENT FOR FREEZING AND DRAINING, AND ACCESSIBILITY OF ALL VALVES, FLANGES, INSTRUMENTS INCLUDING THERMOCOUPLES, ETC.

01000-53-25-020-8120-0841

B

C

D

E

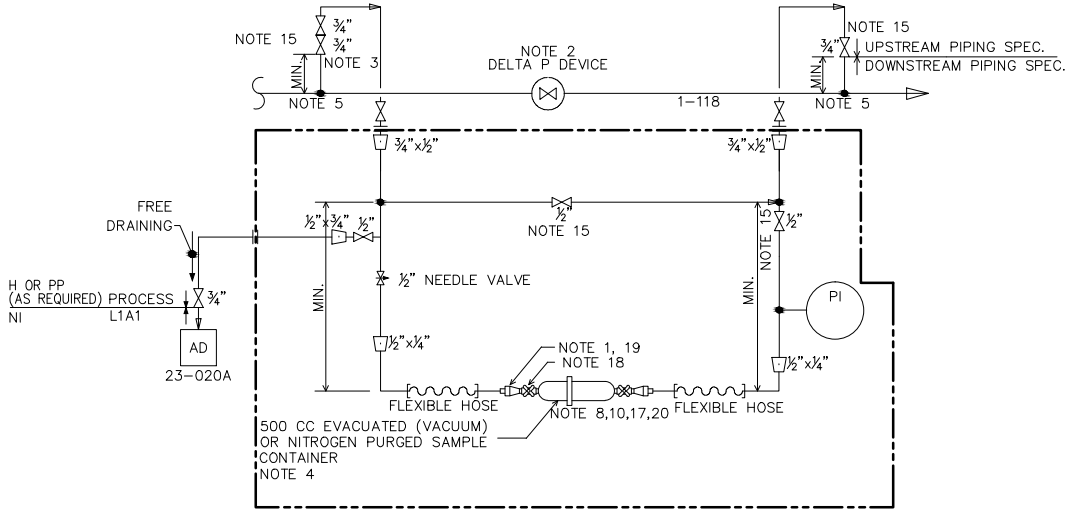
F

G

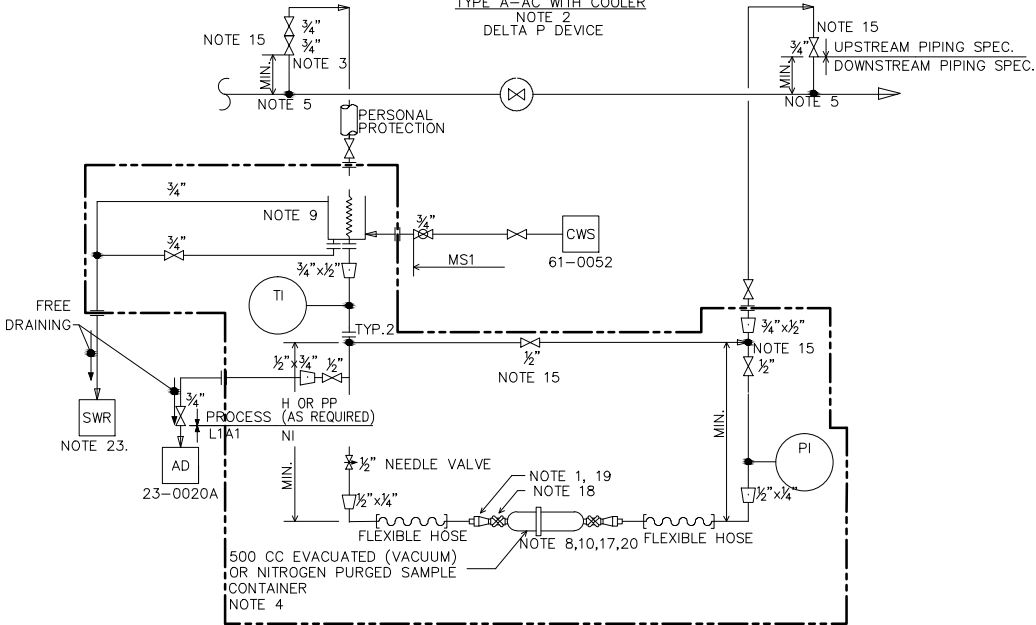
H

J

TYPE A-AC WITHOUT COOLER



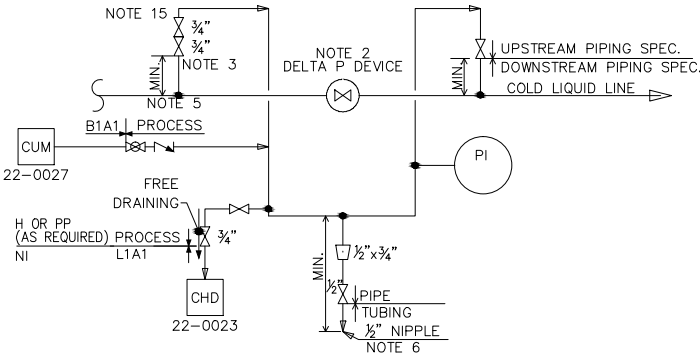
TYPE A-AC WITH COOLER



ITEM NO.	P&ID NO.	SAMPLING LINE NO.	DRAIN LINE NO.	PI	SUMP NO.
SN-2306	14780-8120-25-23-0019	3/4"-CP-23-019014-L2A1-NI	3/4"-AD-23-019015-L2A1/L1A1-NI	23-1907	D-2307
SN-2307	14780-8120-25-23-0020A	3/4"-PL-23-020013-L1A1-NI	3/4"-AD-23-020014-L1A1-NI	23-2006	D-2307

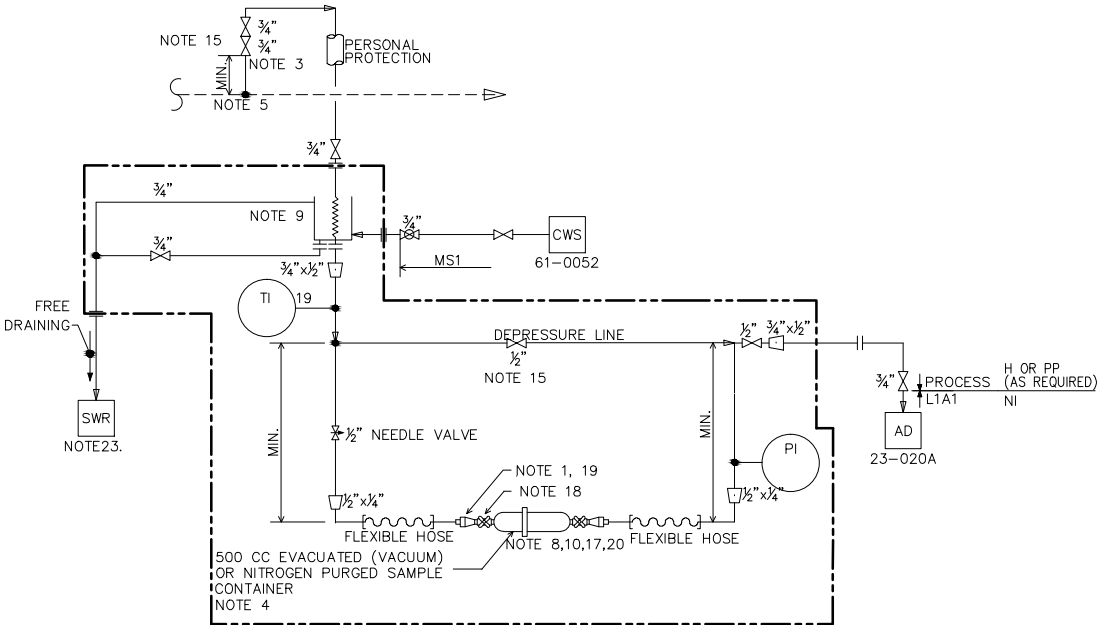
ITEM NO.	P&ID NO.	SAMPLING LINE NO.	DRAIN LINE NO.	COOLANT SUPPLY LINE NO.	COOLANT DRAIN LINE NO.	PI	TI	SUMP NO.
SN-2304	14780-8120-25-23-0014	3/4"-PL-23-014030-L2A1-PP	3/4"-AD-23-014031-L2A1/L1A1-NI/PP	3/4"-CWS-61-052183-MS1-NI	3/4"-SWS-23-014031-MS1-NI	23-1408	23-1406	D-2307
SN-2305	14780-8120-25-23-0017	3/4"-PL-23-017021-L2A1-PP	3/4"-AD-23-017022-L2A1/L1A1-NI/PP	3/4"-CWS-61-052184-MS1-NI	3/4"-SWS-23-017024-MS1-NI	23-1713	23-1707	D-2307

TYPE A-CH WITHOUT COOLER



ITEM NO.	P&ID NO.	SAMPLING LINE NO.	DRAIN LINE NO.	FLUSHING	PI	SUMP NO.
SN-2301	14780-8120-25-23-0008B	3/4"-CUL-23-008046-L1A1-NI	3/4"-CHD-23-008047-L1A1-NI	3/4"-CUL-22-027032-B1A1-NI	23-0813	D-2206
SN-2303	14780-8120-25-23-0009	3/4"-CHP-23-009017-L1A1-NI	3/4"-CHD-23-009018-L1A1-NI	3/4"-CUL-22-027033-B1A1-NI	23-0907	D-2206

TYPE B-AC WITH COOLER



ITEM NO.	P&ID NO.	SAMPLING LINE NO.	DRAIN LINE NO.	COOLANT SUPPLY LINE NO.	COOLANT DRAIN LINE NO.	PI	TI	SUMP NO.
SN-2311	14780-8120-25-23-0020	3/4"-PL-23-020015-L1A1-PP	3/4"-AD-23-020016-L1A1-NI	3/4"-CWS-61-052186-MS1-NI	3/4"-SWS-23-020017-MS1-NI	23-2007	23-2002	D-2307

Z1	AS BUILT	HSO	SWP	JSC	SWL	4/25/16
FS4-3	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	7/1/15
FS4-2	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	6/26/15
FS4-1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	12/29/14
S4	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	10/15/14
S3	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	10/8/14
S2	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	9/16/14
S1	ISSUED FOR CONSTRUCTION WITH HOLDS	HSO	SWP	JSC	SWL	8/20/14
REV.	DESCRIPTION	BY	CHK.	APPR.	PROJ.	DATE

GENERAL NOTES

NOTES

- FOR OPERATING PRESSURE UNDER 125 kg/cm2g USE SWAGELOCK QUICK-CONNECTS. STEMS: SS-QT8-5-8PM BODIES: SS-QT8-8-8PM OR EQUAL FOR OPERATING PRESSURE ABOVE 125 kg/cm2g USE SWAGELOCK QUICK-CONNECTS. STEMS: SS-OT2-5-4PM BODIES: SS-QT2-8-4PM OR EQUAL.
- DELTA P DEVICE IS ANY DEVICE OR INSTRUMENT THAT CREATES A PRESSURE DROP TO ENSURE A SUFFICIENT FLOW THROUGH THE LOOP.
- USE DOUBLE BLOCK FOR ANY CLASS 900# LINES OR HIGHER AND FOR H2 OR H2S SERVICES FOR ANY CLASS 600# LINES OR HIGHER.
- FOR OPERATING PRESSURE UNDER 125 kg/cm2g USE WHITEY 304L-HDF4-500 SAMPLE BOTTLE OR EQUAL FOR OPERATING PRESSURE 125 kg/cm2g AND ABOVE USE XXS 304L PIPE CYCLINDER.
- TAKE-OFF POINTS SHALL BE PLACED ON VERTICAL PIPING OR PLACED ON SIDE OR TOP OF HORIZONTAL PIPING FOR LIQUID SAMPLES AND ON TOP OF HORIZONTAL PIPING FOR GAS SAMPLES.
- BEVEL ALL OPEN ENDS OF SAMPLE CONNECTIONS AT A 45° ANGLE WITH 75 mm MINIMUM EXTENSION.
- USE OIL FLUSH AFTER SAMPLING HEAVY LIQUID.
- ALL SAMPLING BOMBS SHOWN ON THIS PID SHOULD BE EQUIPPED WITH BLOCK VALVES ON EACH SIDE.
- THE COOLING BOX SHALL BE INTERNALLY COVERED WITH "EPOXY" THE COIL SHALL BE TUBING SPEC OR SS.
- A SUPPORT FOR THE CYLINDER "BOOMB" SHALL BE FORESEEN.
- SAMPLING SYSTEM LENGTH MUST BE MINIMUM TO LIMIT THE PRODUCT LOST WHEN FLUSHING.
- ROUTE TO APPROPRIATE FLARE HEADER.
- WHERE SAMPLE COOLER IS REQUIRED THE SUCTION IN TO BE PERSONNEL PROTECTED.
- SAMPLE CONTAINER SHALL BE PROVIDED.
- TYPE OF VALVE IS NOTED IN TABLE, OTHERWISE USE GATE.
- TO USE BALL VALVE.
- SAMPLE CYLINDER, 500CC, 316/316L, 1/4" FEMALE NPT 1800 PSIG.
- SAMPLING VALVE, A182 F316/316L, 3000 PSI 1/4" MALE NPT BOTH ENDS.
- QUICK CONNECTOR, 316SS 300PSI W/INSERT CHECK, 1/4" FEMALE NPT SS-QTM4-B-4PF & SS-QTM4-D-4PF.
- SAMPLE CYLINDER SADDLE, A351 CF8.
- SPECIFIC REQUIREMENT FOR HEAT TRACING AND OBJECTIVE TEMPERATURE SHALL BE AS INDICATED AT SAMPLE POINT LOCATION ON THE PIPING AND INSTRUMENT DIAGRAM. WHEN HEAT TRACING IS REQUIRED THE CLOSED SAMPLING SYSTEM SHALL INCLUDE A HEATED ENCLOSURE.
- FOR SAMPLE TYPE A-FR AND B-FR IF TRACING TEMPERATURE IS 50 USE EMPERED COOLING WATER FOR SAMPLE COOLER.
- DESTINATION OF LINE TO BE ROUTED TO TRENCH INSTEAD OF FUNNEL.



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uop BASED ON

PIPING AND INSTRUMENT DIAGRAM
SAMPLE CONNECTION TYPES (1/2)
CONCENTRATION, DECOMPOSITION AND NEUTRALIZATION
SECTION UOP PHENOL PROCESS UNIT

BY	DATE	DWG.	DSGN.	CHK.	APPR.
DATE	DATE				
PROJ. NO.	DWG. NO.				REV.
PEN	120117				Z1
PPCL	14780-8120-25-23-0001D				

SCALE